

“Car Rental System”

Dept. of Computer Science and Technology

A PROJECT ON CAR RENTAL SYSTEM USING OBJECT ORIENTED PROGRAM



Software Development Program - I

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Abstract

The Car Rental System is developed to serve the Car Rental service shop owners to make their service computerized and easier to handle for them to serve their customer. Our program is very much simple and user friendly so that the Car Rental service shop owners can easily operate it. It comes with password protection facility so that the owners can keep their file safe.

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Declaration

We hereby declare that the Project on Car Rental System submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering of Bangladesh University of Business and Technology (BUBT) is our own work and that it contains no material which has been accepted for the award to the candidate(s) of any other degree or diploma, except where due reference is made in the text of the project. To the best of our knowledge, it contains no materials previously published or written by any other person except where due reference is made in the project.

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Chapter 1

Introduction

1.1 Introduction

Computers have two main parts: hardware and software like vocal cord (hard-ware) and lyrics (software). In this section: hardware the computer is an amazingly useful general-purpose technology, to the point that now cameras, phones, thermostats, and more are all now little computers. This section will introduce major parts and themes of how computer hardware works. Hardware refers to the physical parts of the computer, and software refers to the programmed code that runs on the computer. Car Rental System is a project which helps the owner of the Car Rental Service shop to provide their customers digitally so that they don't have to depend on analog system. As it is a computerized system, it will create transparency to the customers.

1.2 Objectives

This objective of this project is to provide software based rental system to make the service computerized. We have given many facilities for users in this project.

- 1) Rent a car Interface
- 2) Password change
- 3) Addition of new car
- 4) Rent history check

1.3 Project Scope

The scope of this project Car Rental System is to develop an OOP based software to support detailed outcome.

This Program will be very useful for the computers with a very nil processing power and very little RAM (Random Access Memory).

Processors: Atom, Celeron (Dual Core)

RAM: 2 GB.

1.4 Our Contributions

It is easy to handle and operate on low end devices.

This can be operate easily on old operating system such as windows 7, windows xp and linux.

1.5 Conclusions

Nothing is perfect in this world. So we are also no exceptions. Although, we have tried our best to present the information effectively, yet there can be further enhancement of the application. We have taken care of all the critical aspects, which need to take care of during the development of project. Like the things this project also has some limitations and can further be enhanced by someone because there are certain drawbacks that do not permit the system 100 percent accurate.

Chapter 2

Existing Literature

2.1 Introduction

A car rental service is a service that serves its customers with a scope to rent a car by their choice. It is quite popular in recent day because of

People can choose what car they want to travel or go wherever they want. Human always like to discover the new things. Travelling to a new place can give us the scope to do more discoveries. Car rental services can provide them to travel and keep their mind and body charm and sound.

The concept of car rental has gained a significant attention in recent years, driven by the increasing demand for flexible and convenient transportations solutions. It can be the scope to gain the popularity among the people by using it properly.

2.2 Historical Overview

A car rental is a company that rents automobile for a short period of time, generally ranging from few hours to few weeks. Car rental agencies primarilly serve those people who require a temporary

vehicle, for example, those who do not have their own car, travelers who are out of the town, or owner of damaged or destroyed vehicle who are awaiting repair or insurance compensation.

The earliest known example of cars being offered for rent dates back to 1906. The German company Sixt was established in 1912 under the name Sixt Autofahrten und Selbstfahrer (Sixt Car Cruises and Self Drivers) Joe Saunders of Omaha, Nebraska first started with only one borrowed Model T Ford in 1916, but by 1917, his Ford Livery Company was renting out 18 Model Ts at 10 cents per mile. The company name because Saunderson *Drive-It-Yourself* system and then Saunders system. By 1926, Saunders had expanded to 56 cities. Saunders' company was bought by Avis in 1955. An early competitor to Saunderson was Walter L. Jacobs, whose Chicago-based *Rent-a-Car* opened in 1918 with twelve Ford Model T. The company was bought in 1923 by John Hertz. In Britain, car rental started with Godfrey Davis, established in 1920, and bought by Europcar in 1981 (Walter L. Jacobs, 88: *Rent-a-Car Pioneer*, 2015). The sector expanded rapidly in the U.S. in 1926, the American Driveursel Association assembled over 1200 delegates in Chicago. The growth in travel after World War II led to the establishment of several well-known international companies, including National Car Rental (1947), Europcar (1949), Enterprise *Rent-a-Car* (1957), Thrifty Rent A Car (1958), and Budget Rent a Car (1958), (Walter L. Jacobs, 88: *Rent-a-Car Pioneer*, 2015).

2.3 Car Rental Service In Bangladesh

According to industry sources, there are now around 1000 car rental groups operating in the country, up from around 100 ten years ago. They range in size from companies with just a handful of vehicles to larger groups with a fleet of several hundred cars and buses. For the past few years the car hire market has been showing double-digit growth.

Currently, the biggest clients are the mobile telecom operators, multinational companies and non-governmental organizations (NGOs).

Dhaka Tours Rent-A-Car Association, one of the oldest car rental companies in the city, has 50 minibuses and cars (Sedans), which they rent out to Grameen Phone, Nokia, Siemens, Bangladesh Bank, Dhaka Sheraton and Pan Pacific Sonargaon Hotel.

2.4 Conclusions

The Car Rental Service has brought a revolutionary change in car industry. People are more interested in car rental service due to time save and also the save of their money.

Chapter 3

System Architecture Model

3.1 Introduction

1. The existing system is manual.
2. In this system the user will be able to rent a car for his customer.
3. This system is pretty much user friendly and easy to operate.
4. Password of the user is maintained in car pass.txt.

3.2 Software Requirement

Platform Used: Windows or Linux, Codeblocks (IDE) with GNU GCC Compiler.

Hardware Specification:

*RAM: 2 GB.

*Processor: Atom or higher (Minimum 1 GHz Speed)

3.3 Methodology

To Implement this program we need some concepts that we will use to develop this program. We will develop this code using Object Oriented Program.

Object-oriented programming (OOP) is a computer programming model that organizes software design around data, or objects, rather than functions and logic. An object can be defined as a data field that has unique attributes and behavior.

OOP focuses on the objects that developers want to manipulate rather than the logic required to manipulate them. This approach to programming is well-suited for programs that are large, complex and actively updated or maintained. This includes programs for manufacturing and design, as well as mobile applications; for example, OOP can be used for manufacturing system simulation software.

Concepts that are used to develop the program:

1. Class and Objects
2. Conditional Statements
3. File Handling
4. Iteration/Loop
5. Exception Handling

6. User Authentification

7. Functionality Decomposition

8. Data Validation

9. Password Mangement

Chapter 4

Implementation of Our System

4.1 Introduction

Explanation of the key Functions:

1. Main Menu:

In this user will see the options where he can choose his desired option.

2. Car Rent Menu:

In this user will put his password and customer's name and select a car the customer want to rent.

3. Change pass Menu:

In this user can change his password.

4. Adding New Car Menu:

In this user can add new car to this program just writing the car name and its rate.

5. Rent History:

In this user can see the rental history that whom he rented the car and the name of the car and calculated total money.

6. Developer Info:

In this user can see who developed this program.

7. Exit:

In this user will be able to exit from the program whenever he wants to terminate his work.

4.2 overview

Here is the graphical presentation of the project:



Figure 4.1 : The graphical presentation

Chapter 5

Experimental Result and Evaluation

5.1 Introduction

Here we have analyzed all the results regarding to the project and also we have tried to show the output interfaces of our program.

5.2 Result Analysis

Some screenshots of our program output are shown below in order.
Starting With the Main Menu.

```
std::cout<< "\t\t\tMain Menu\n"<<std::endl; ///showing main menu options
std::cout << "\t\t\t1. Rent a Car\n";
std::cout << "\t\t\t2. Change Password\n";
std::cout<< "\t\t\t3. Add New Car\n";
std::cout<< "\t\t\t4. Rent History\n";

std::cout<< "\t\t\t5. Developer info\n";
std::cout << "\t\t\t6. Exit\n\n";
std::cout << "\t\t\tEnter your choice: \n";
std::cout<<"\t\t\t";
std::cin >>option;
```

Figure 5.1: SS of Main Menu

```
Welcome to the system.

Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
```

Figure 5.2: Main Menu

```
Enter your choice:  
1  
Enter password:****  
Access Granted  
Enter Customer's name: Rafi
```

Figure 5.3: Rent Car Interface

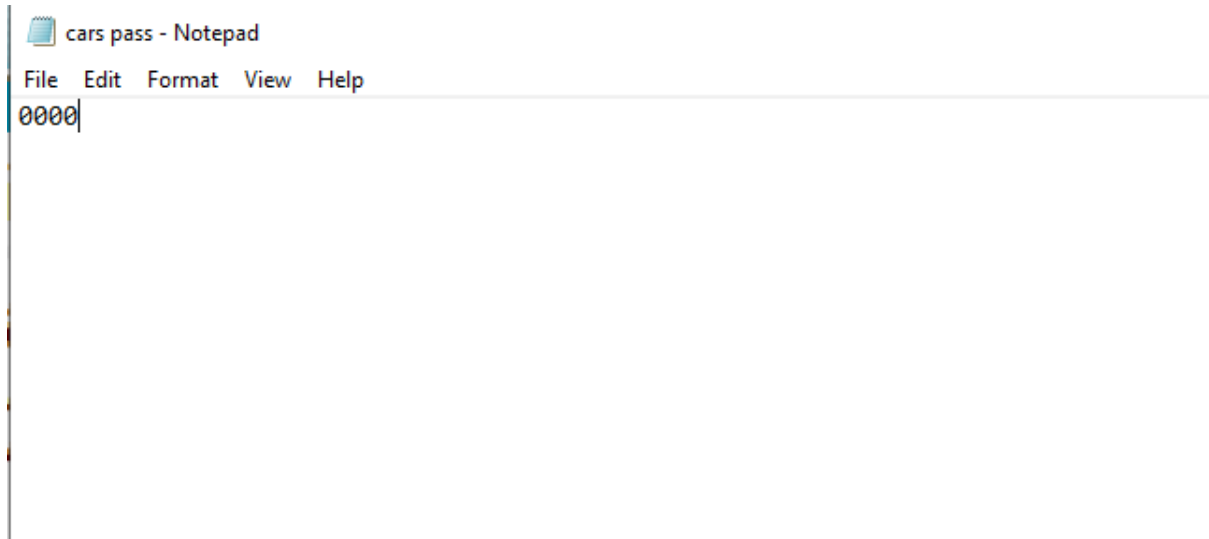


Figure 5.4: File Read to Check Password


```

if (choice >= 1 && choice < count)
{
    file.clear();///
    file.seekg(0, std::ios::beg);///
    for (int i = 0; i < choice; i++)
    {
        file >> carName >> dailyRate;
    }

    std::cout << "\n\t\t\tNumber of days you want the car(At least 1 Day): ";
    do
    {
        std::cin >> days;
        if (days <= 0)
        {
            //std::cout << "\t\t\tNegative number can't be taken!"<<std::endl;
            std::cout<<"\t\t\tEnter a valid number:";
            //std::cout<<"\t\t\t";
        }
    }
    while (days <= 0);
}

```

Figure 5.7: SS of Code for Car Choosing and Selecting Days

```

Enter the number of your choice: 3
Number of days you want the car(At least 1 Day): 3

```

Figure 5.8: Entering the Desired Car and Rental Days


```

double calculateTotal()
{
    double total = car.dailyRate * days;
    if (total >= 1500.0)
    {
        discount = 0.05 * total;
        total -= discount;
    }
    return total;
}

void displayInvoice() ///method for display the invoice
{
    std::cout << "\n\t\t\tCar Rental System - Customer Invoice\n";
    std::cout << "\t\t\tCustomer name: " << customerName << "\n";
    std::cout << "\t\t\tCar: " << car.name << "\n";
    std::cout << "\t\t\tNumber of days: " << days << "\n";
    std::cout<< "\t\t\tTotal:"<<(car.dailyRate*days)<<" $\n";
    std::cout << "\t\t\tTotal Rent After Discount:" << calculateTotal() << " $\n";
    std::cout << "\t\t\tTotal Discount: " << discount << " $\n";
    std::cout<<std::endl;
    std::cout<<std::endl;
}

```

Figure 5.9: SS of Code for Calculate Total and Invoice

```

Car Rental System - Customer Invoice
Customer name: Rafi
Car: Lancer
Number of days: 3
Total:225 $
Total Rent After Discount:225 $
Total Discount: 0 $

```

Figure 5.10: Invoice print

```

std::string EnterPassword()
{
    std::string inputPassword; ///inputting password from user
    char ch;
    std::cout<<"\t\t\tEnter password:";
    while (true)
    {
        ch = _getch();
        if (ch == 13) /// ASCII value of Enter key
            break;
        else if (ch == 8) ///ASCII value of Backspace key
        {
            if (inputPassword.length() > 0)
            {
                std::cout << "\b \b";
                inputPassword.pop_back();
            }
        }
        else
        {
            std::cout << '*';
            inputPassword.push_back(ch); ///entering asterisk instead of hard value
        }
    }
    std::cout << std::endl;
    return inputPassword;
}

```

Figure 5.11: SS of Code to Take Password Input without Display

```

Enter your choice:
1
Enter password:****

```

Figure 5.12: Password Input

```
bool Password(std::string &pass)
{
    std::ifstream filex("cars pass.txt");
    if (!filex.is_open())
    {
        std::cout << "Error finding file\n";
        return false;
    }
    else
    {
        std::string storedPassword; ///stored password will be stored in this variable
        std::getline(filex >> std::ws, storedPassword); ///using whitespace
        std::string inputPassword = EnterPassword(); ///passing the input pass into enterpassword method
        return PasswordMatch(inputPassword, storedPassword);
    }
}
```

Figure 5.13: SS of Code to Read Password from Text File

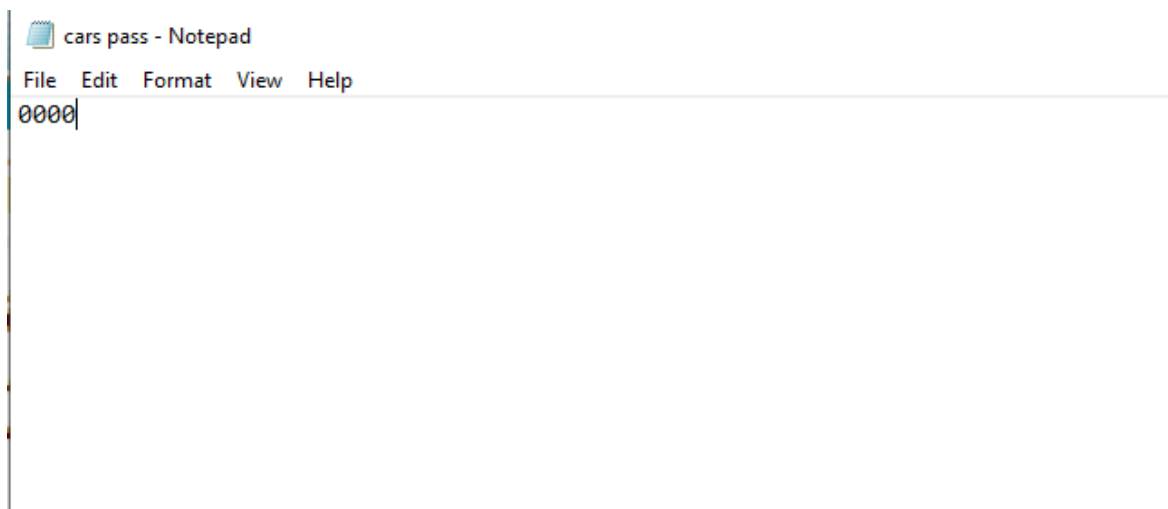


Figure 5.14: Text File Containing Password

```

bool PasswordMatch(const std::string &inputPassword, const std::string &storedPassword)
{
    if (inputPassword == storedPassword)
    {
        std::cout << "\t\t\tAccess Granted" << std::endl;
        return true;
    }
    else
    {
        std::cout << "\t\t\tPassword didn't match" << std::endl;
        return false;
    }
}

```

Figure 5.15: SS of Code for matching the input and stored password

```

void displayWelcomeMessage(const std::string &messageFile)
{
    std::ifstream file(messageFile);
    if (!file)
    {
        throw std::runtime_error("Can't open message file: " + messageFile);
    }

    std::string line;
    while (std::getline(file, line))
    {
        std::cout << "\t\t\t" << line << std::endl;
    }
    std::cout << "\n\n";
}

```

Figure 5.16: SS of Code to Show Welcome Message

```

void changePassword(std::string &password)
{
    std::ifstream filey("cars pass.txt"); ///reading data from car
    if (!filey.is_open())
    {
        throw std::runtime_error("Error finding file");
        return;
    }

    std::string storedpass;
    std::getline(filey >> std::ws, storedpass); ///using whitespace
    filey.close();

    std::string inputPassword;
    bool passwordMatches = false;

```

Figure 5.17: SS of Code for Change Password (1)

"C:\Users\pc\Downloads\program c++\class test 02\SDP - I\New folder (2)\car rental final resolved test.exe"

Welcome to the system.

Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:

2

Enter password:****

Figure 5.18: Password Change Menu

```

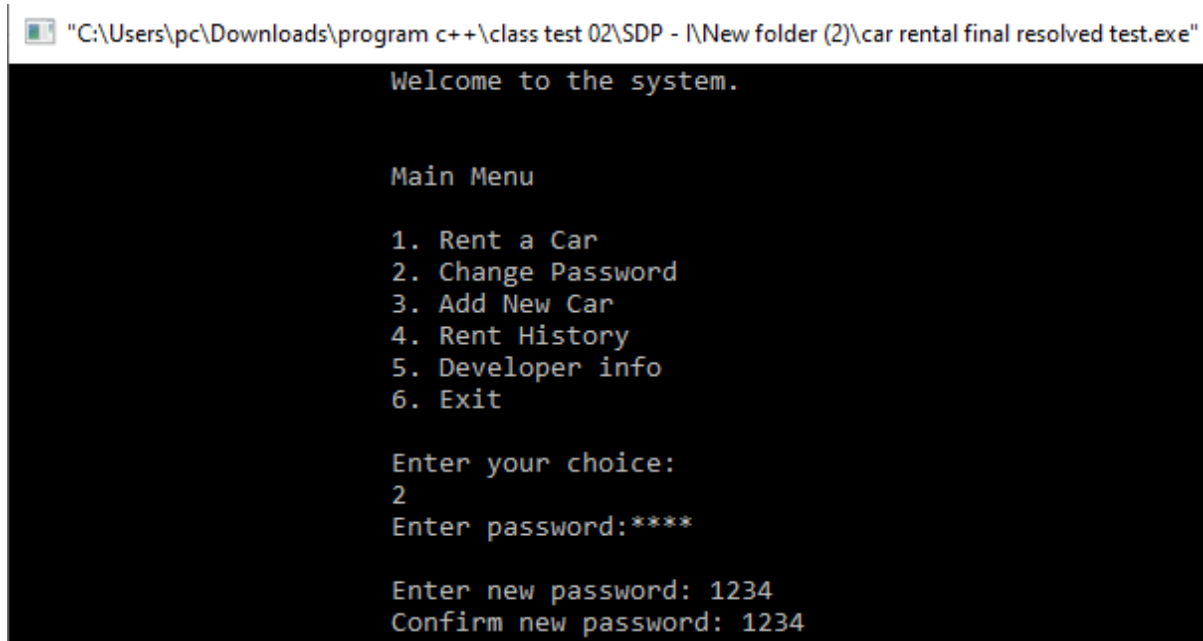
do
{
    inputPassword = EnterPassword();
    if (inputPassword == storedpass)
    {
        std::string newpass, confirmpass;
        bool passwordsMatch = false;
        do
        {
            std::cout << "\n\t\t\tEnter new password: ";
            std::cin >> newpass;
            std::cout << "\t\t\tConfirm new password: ";
            std::cin >> confirmpass;

            if (newpass == confirmpass)
            {
                std::ofstream filez("cars pass.txt"); ///writing new pass
                if (!filez.is_open())
                {
                    throw std::runtime_error("Error finding file");
                    return;
                }

                std::string tempPassword = newpass;
                std::cout << "\t\t\tPassword successfully changed" << std::endl;
                filez << tempPassword;
            }
        }
    }
}

```

Figure 5.19: SS of Code for Password Change (2)



```

"C:\Users\pc\Downloads\program c++\class test 02\SDP - I\New folder (2)\car rental final resolved test.exe"

Welcome to the system.

Main Menu

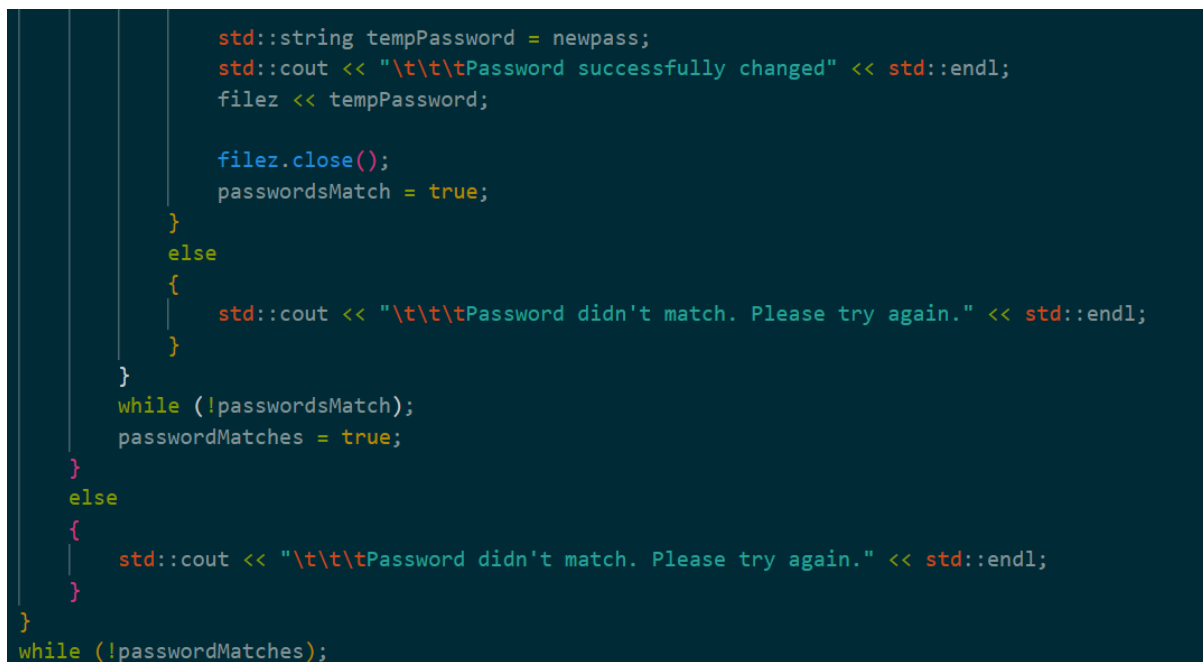
1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
2
Enter password:****

Enter new password: 1234
Confirm new password: 1234

```

Figure 5.20: Comparing New Entered Password



```

std::string tempPassword = newpass;
std::cout << "\t\t\tPassword successfully changed" << std::endl;
filez << tempPassword;

filez.close();
passwordsMatch = true;
}
else
{
    std::cout << "\t\t\tPassword didn't match. Please try again." << std::endl;
}
}
while (!passwordsMatch);
passwordMatches = true;
}
else
{
    std::cout << "\t\t\tPassword didn't match. Please try again." << std::endl;
}
}
while (!passwordMatches);

```

Figure 5.21: SS of Code for Password Change (3)

"C:\Users\pc\Downloads\program c++\class test 02\SDP - I\New folder (2)\car rental final resolved test.exe"

```
Welcome to the system.

Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
2
Enter password:****

Enter new password: 1234
Confirm new password: 1234
Password successfully changed
```

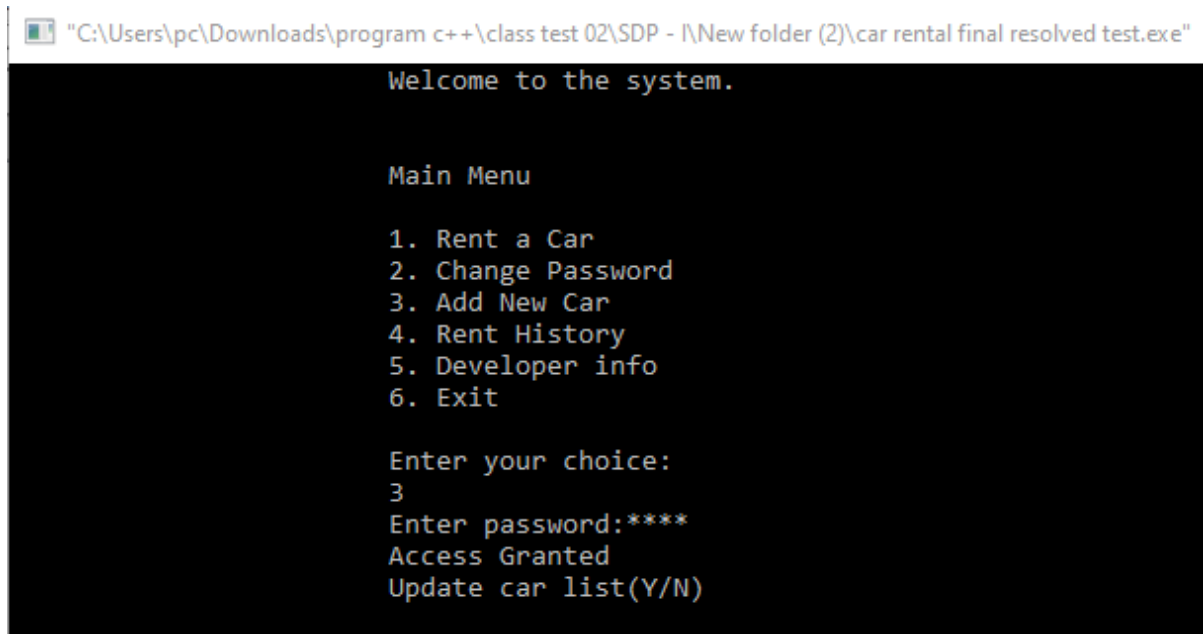
Figure 5.22: Password Changed Menu


```

void Update_car_status(const std::string &carlist)
{
    std::string opt;
    std::cout<<"\t\t\tUpdate car list(Y/N)"<<std::endl;
    std::cout<<"\t\t\t";
    std::cin>>opt;
    if(opt=="Y")
    {
        std::ifstream fileA(carlist);
        if(!fileA.is_open())
        {
            throw std::runtime_error("File can't be found"+carlist);
        }
        else
        {
            std::string carname;
            double rate;
            while(fileA>>carname>>rate)
            {
                std::cout<<"\t\t\tCar Name: "<<carname<<" " Rate: "<<rate<<"$"<<std::endl;
            }
            fileA.close();
            std::ofstream fileB(carlist,std::ios::app); ///appends with the previous file
            if(!fileB.is_open())
            {
                throw std::runtime_error("File can't be found!");
            }
        }
    }
}

```

Figure 5.23: SS of Code for Adding New Car (1)



```

"C:\Users\pc\Downloads\program c++\class test 02\SDP - \New folder (2)\car rental final resolved test.exe"

Welcome to the system.

Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
3
Enter password:****
Access Granted
Update car list(Y/N)

```

Figure 5.24: Add New Car Menu

```
Update car list(Y/N)
Y
Car Name: Premio    Rate: 56$
Car Name: Corolla   Rate: 60$
Car Name: Lancer    Rate: 75$
Car Name: Axio      Rate: 80$
Car Name: Noah      Rate: 75.5$
Car Name: Hiace     Rate: 65.6$
Car Name: Probox    Rate: 40.5$
Car Name: Allion    Rate: 56$
Car Name: Superior  Rate: 78.5$
Car Name: harrier   Rate: 150$
Car Name: Tesla     Rate: 300$
Car Name: proton    Rate: 90.5$
Car Name: Sonata    Rate: 95.5$
Car Name: civic     Rate: 120$
```

Figure 5.25: Showing Previous Added Cars

```

fileA.close();
std::ofstream fileB(carlist,std::ios::app); ///appends with the previous file
if(!fileB.is_open())
{
    throw std::runtime_error("File can't be found!");
}
else
{
    std::string newcar;
    std::cout<<"\t\t\tEnter new car:\t";
    std::cin>>newcar;
    double newrate;
    std::cout<<"\t\t\tEnter rate of the car:\t";
    std::cin>>newrate;
    fileB<<newcar<<" "<<newrate<<std::endl;
    fileB.close();
}
}
else
{
    std::cout<<"\t\t\tOK\n"<<std::endl;
}
}

```

Figure 5.26: SS of Code for Adding New Car (2)

```

Enter new car:  Fielder
Enter rate of the car:  60.5

```

Figure 5.27: Adding New Car and Rate

```

std::ifstream fileC(carlist);
if(!fileC.is_open())
{
    throw std::runtime_error("File can't be found");
}
std::string carname1;
double rate1;
while(fileC>>carname1>>rate1)
{
    std::cout<<"\t\t\tCar Name: "<<carname1<<" "<<"   Rate: "<<rate1<<"$"<<std::endl;
}
fileC.close();
std::cout<<"\n\n";

```

Figure 5.28: SS of Code for Adding New Car (3)

```

Car Name: Premio    Rate: 56$
Car Name: Corolla   Rate: 60$
Car Name: Lancer    Rate: 75$
Car Name: Axio      Rate: 80$
Car Name: Noah      Rate: 75.5$
Car Name: Hiace     Rate: 65.6$
Car Name: Probox    Rate: 40.5$
Car Name: Allion    Rate: 56$
Car Name: Superior  Rate: 78.5$
Car Name: harrier   Rate: 150$
Car Name: Tesla     Rate: 300$
Car Name: proton    Rate: 90.5$
Car Name: Sonata    Rate: 95.5$
Car Name: civic     Rate: 120$
Car Name: Fielder   Rate: 60.5$

```

Figure 5.29: Updated Car List with New Car

```
else if(option=="3") ///car update
{
    bool passwordEntered="false";
    do
    {
        passwordEntered=Password(inputPassword);
    }
    while(!passwordEntered);
    Update_car_status("cars.txt");
}
```

Figure 5.30: SS of Code for Adding New Car in the main() function

```

void Invoicehistory(const std::string &Memo) ///method for showing invoice history menu
{
    std::ifstream fileIN(Memo); ///using ifstream to read file
    if(!fileIN.is_open())
    {
        throw std::runtime_error("File can't be found"+Memo);
    }
    else
    {
        while(fileIN.good())
        {
            std::string Output;
            getline(fileIN,Output);
            std::cout<<"\t\t\t"<<Output<<std::endl;
        }
    }
    fileIN.close();
}

```

Figure 5.31: SS of Code to Print Rent History

```

else if(option=="4") ///invoice history
{
    bool passwordEntered="false";
    do
    {
        passwordEntered=Password(inputPassword);
    }
    while(!passwordEntered);

    Invoicehistory("Invoice.txt");
}

```

Figure 5.32: SS of Code to Print Rent History in the main() Function

"C:\Users\pc\Downloads\program c++\class test 02\SDP - I\New folder (2)\car rental final resolved test.exe"

```
Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
4
Enter password:****
Access Granted

Name:hasib      Car:Lancer      Days:3  Total($):225  Discount($):0
Name:Eshika     Car:harrier     Days:2  Total($):300  Discount($):0
Name:Rafi       Car:Lancer      Days:3  Total($):225  Discount($):0
```

Figure 5.33: Rent History Menu

```

void developer_info(const std::string &info) ///method for
{
    std::ifstream filei(info);
    if(!filei.is_open())
    {
        throw std::runtime_error("File can't be found");
    }
    std::string line,n1,n2,n3;
    while(std::getline(filei,line))
    {

        std::cout<<"\t\t\t"<<line<<std::endl;

    }
    filei.close();
    std::cout<<std::endl;
}

```

Figure 5.34: SS of Code for Showing Developer Info

```

else if(option == "5")///showing developer info
{
    developer_info("developer_info.txt");
    // break;
}

```

Figure 5.35: SS of Code for Showing Developer Info in main()

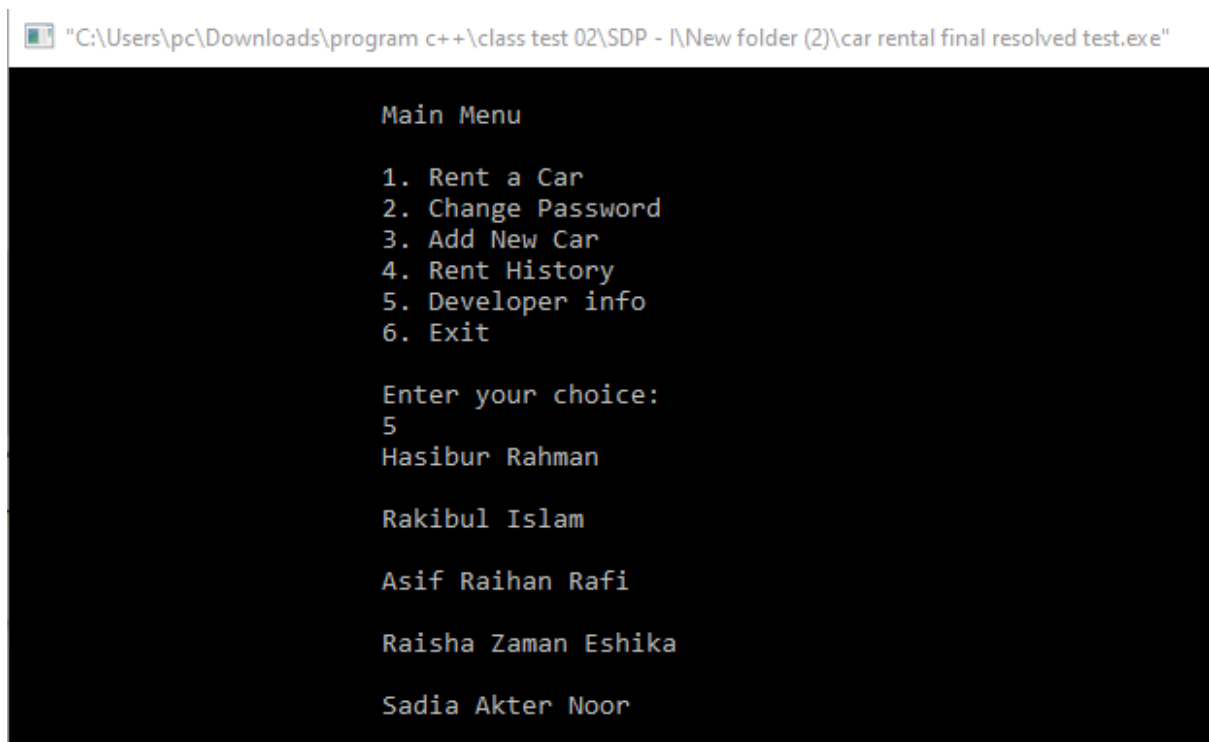
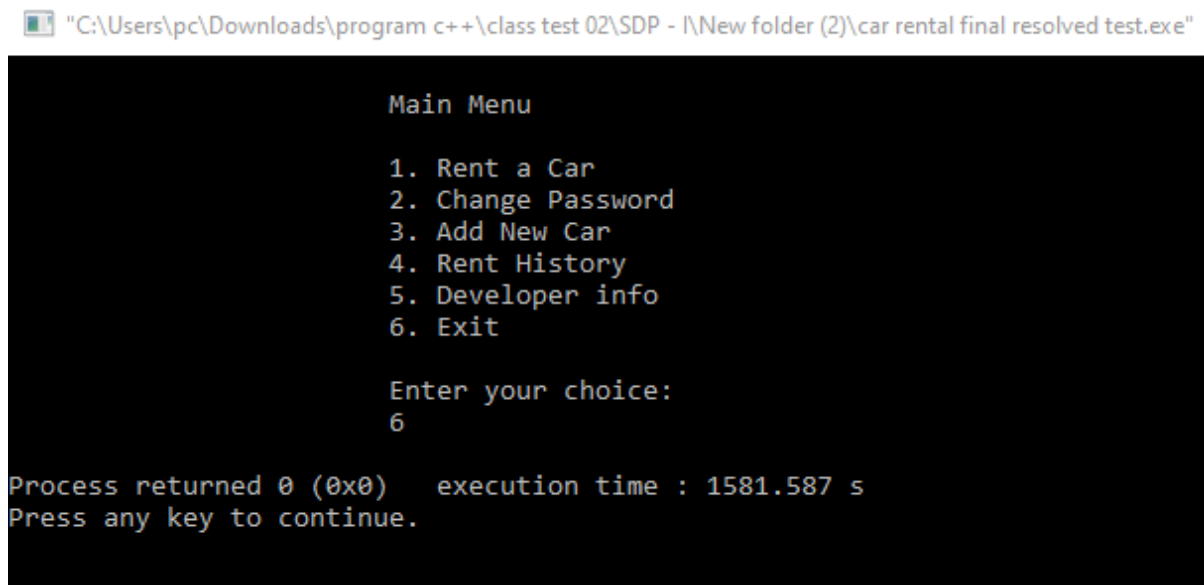


Figure 3.36: Developer Info Menu

```
else if (option == "6")
{
    break;
}
```

Figure 3.37: SS of Code for Exiting the Program



```
"C:\Users\pc\Downloads\program c++\class test 02\SDP - I\New folder (2)\car rental final resolved test.exe"

Main Menu

1. Rent a Car
2. Change Password
3. Add New Car
4. Rent History
5. Developer info
6. Exit

Enter your choice:
6

Process returned 0 (0x0)   execution time : 1581.587 s
Press any key to continue.
```

Figure 5.38: Exit Menu

Chapter 6

Future Scope and Limitation

6.1 Future Scope of Our Project

In future we would like to add more enhancement password protection, multi-user support facility, error handling refinement, additional features like car tracking, classifications of cars, make this into online platform where people can rent car from home by using Internet.

Error Handling Refinement:

We would like to improve error handling by implementing robust mechanism to handle various exceptional scenarios, ensuring smooth program execution.

Security Enhancement:

We would like strengthen the security of this program by incorporating encryption techniques to protect sensitive data, such as customer's information, user's passwords etc.

Additional Features:

We would like to introduce some additional functionalities, such as rental reservation, car availability tracking, customer feedback collection and classifications of cars as SUV, Sedan, minibus etc.

Multi-user Support:

We would like to extend the application to support multiple users simultaneously, enabling rental agents to manage bookings and customers effectively.

Penalty System:

We would like to add a penalty system for those who won't return the car in time.

6.2 Limitation

The limitation of our project is very nil because it is meant to be like this.

6.3 Conclusions

After designing this whole project using Object Oriented Programming we can say that even better interfaces and output can be obtained from here with a very little and hassle free modifications. But our journey has just begun and there is long way to go and a world we have in our hand to learn.

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